### Relation to Toroidal Graphs

Consider a closed curve $C$ on the punctured torus with $n$ self-intersections. It can be realized as a directed toroidal graph (graph that can be embedded on a torus) with certain conditions.

- It must have $n$ vertices.
- For each vertex $v_i$, $\deg^+(v_i) = 2$ and and $\deg^-(v_i) = 2$.
- There are $\frac{4n}{2} = 2n$ edges (four for each vertex and then each edge is counted twice)
- It should be strongly connected (there is a directed path between any two vertices).

Let this graph be $G_C$. Let $k_C$ be the maximum length of ac directed simple path (path which does not repeat vertices) in $G_C$. It is clear that $k_C \leqslant n - 1$.

### Simple Subsets

Fix a closed curve $C$ on the punctured torus with $n$ self-intersections. Consider simple path-connected subsets of the image of $C$, $\gamma$ and define the length of a $\gamma$ to be $L(\gamma)$ = number of self-intersection points that lie on $\gamma$. Let $k_C = \max_\gamma L(\gamma)$ where $\gamma$ ranges over all such subsets of the image of $C$. It's clear that $k_C \leqslant n$.

## ABSTRACT

The fundamental group $\pi_1(S)$ of a surface $S$ with boundary describes the closed curves on the surface up to base homotopy, or continuous deformation. The mapping class group describes the symmetries of the surface, but also acts on the fundamental group. In a recent paper, Maryam Mirzakhani describes the growth of the number of orbits of this action for closed curves up to a certain hyperbolic length $L$ and at most $k$ self-intersections, as $L \to \infty$.

We work on the punctured torus and the 4-holed sphere. We present an exhaustive method for classifying the orbits of this action on free homotopy classes with a given minimal self-intersection number $k$, using the method of basepoints found by Crisp, et al. in the 1990s. We show the classification for $n = 4$ on the punctured torus and discuss how the method can be easily extended for greater $n$. We also present a way of utilizing basepoints to study orbits on the 4-holed sphere using planar graphs. During the project, we wrote a library of Java code to algorithmically perform this classification on the torus using word representations of free homotopy classes as elements of a free group on two generators $F(a, b)$.